

Double-click (or enter) to edit

House Price Prediction using Linear Regression

Importing Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split

print("Libraries Imported Successfully")
```

Libraries Imported Successfully

Loading the Dataset

```
url = "https://raw.githubusercontent.com/selva86/datasets/master/BostonHousing.csv"

data = pd.read_csv(url)

print(data.head())
```

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	\
0	0.00632	18.0	2.31	0	0.538	6.575	65.2	4.0900	1	296	15.3	
1	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	242	17.8	
2	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242	17.8	
3	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222	18.7	
4	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222	18.7	

	b	lstat	medv
0	396.90	4.98	24.0
1	396.90	9.14	21.6
2	392.83	4.03	34.7
3	394.63	2.94	33.4
4	396.90	5.33	36.2

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Understanding the Dataset

```
print(data.shape)
```

(506, 14)

```
print(data.columns)
```

```
Index(['crim', 'zn', 'indus', 'chas', 'nox', 'rm', 'age', 'dis', 'rad', 'tax',
      'ptratio', 'b', 'lstat', 'medv'],
      dtype='object')
```

Preparing Input and Output Data

```
X = data.drop('medv', axis=1)

y = data['medv']

print(X.head())

print(y.head())
```

```

      crim    zn  indus  chas    nox    rm    age    dis  rad  tax  ptratio  \
0  0.00632  18.0   2.31    0  0.538  6.575  65.2  4.0900  1  296    15.3
1  0.02731   0.0   7.07    0  0.469  6.421  78.9  4.9671  2  242    17.8
2  0.02729   0.0   7.07    0  0.469  7.185  61.1  4.9671  2  242    17.8
3  0.03237   0.0   2.18    0  0.458  6.998  45.8  6.0622  3  222    18.7
4  0.06905   0.0   2.18    0  0.458  7.147  54.2  6.0622  3  222    18.7

      b    lstat
0  396.90    4.98
1  396.90    9.14
2  392.83    4.03
3  394.63    2.94
4  396.90    5.33
0    24.0
1    21.6
2    34.7
3    33.4
4    36.2
Name: medv, dtype: float64

```

✓ Splitting the Dataset

```

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

```

```

print("Training Data Shape:", X_train.shape)
print("Testing Data Shape:", X_test.shape)

```

```

Training Data Shape: (404, 13)
Testing Data Shape: (102, 13)

```

✓ Training the Linear Regression Model

```

model = LinearRegression()

model.fit(X_train, y_train)

print("Model Trained Successfully")

```

```

Model Trained Successfully

```

✓ Predicting House Prices

```

y_pred = model.predict(X_test)

print(y_pred[:5])

```

```

[28.99672362 36.02556534 14.81694405 25.03197915 18.76987992]

```

✓ Comparing Actual and Predicted Prices

```

comparison = pd.DataFrame({
    'Actual Price': y_test,
    'Predicted Price': y_pred
})

print(comparison.head(10))

```

```

      Actual Price  Predicted Price
173          23.6      28.996724
274          32.4      36.025565
491          13.6      14.816944
72           22.8      25.031979
452          16.1      18.769880
76           20.0      23.254429
316          17.8      17.662538
140          14.0      14.341190
471          19.6      23.013207
500          16.8      20.632456

```

▼ Data Visualization

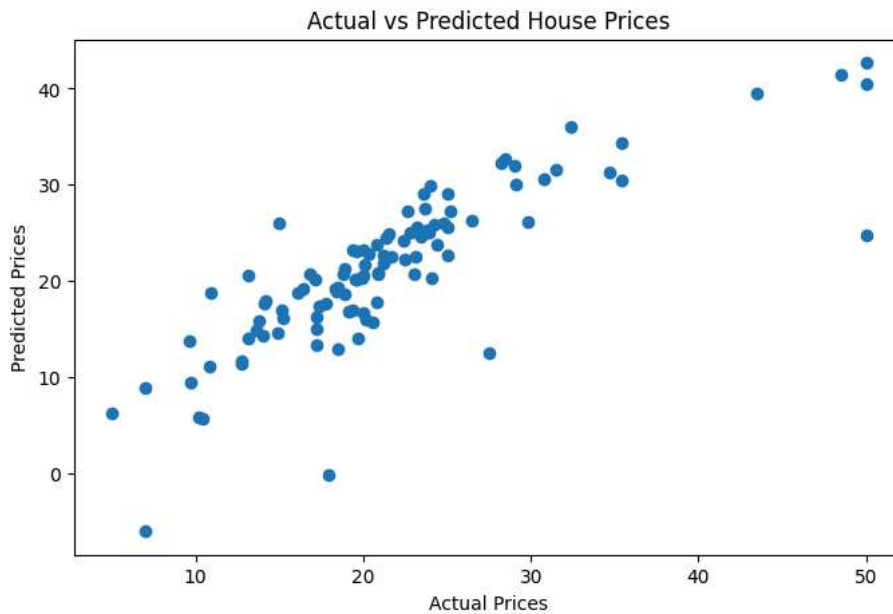
```
plt.figure(figsize=(8,5))

plt.scatter(y_test, y_pred)

plt.xlabel("Actual Prices")
plt.ylabel("Predicted Prices")

plt.title("Actual vs Predicted House Prices")

plt.show()
```



▼ Model Accuracy

```
from sklearn.metrics import r2_score

score = r2_score(y_test, y_pred)

print("R2 Score:", score)
```

R2 Score: 0.6687594935356326